

Course : Artificial Intelligence

Course code : PCA2207

QP code : 642207

UNITWISE QUESTION BANK

UNIT - 1

1. Define artificial intelligence.
2. What are the types of heuristic functions?
3. Define the term agent.
4. What is sensors and effectors?
5. What are the characteristics of a fully observable environment?
6. Define uninformed and informed search.
7. Write the meaning of syntax and semantics.
8. Write the applications of AI.
9. Discuss the composed of artificial intelligence.
10. Explain informed search algorithm.
11. Explain the concept of heuristic search and its advantages over uninformed search.
12. Explain the structure of intelligent agents, including the different types of agent architecture.
13. Discuss the different types of environments and their implications for agent design.
14. Compare and contrast online and offline search agents, including their applications and limitations.

UNIT – 2

1. What is a constraint satisfaction problem(CSP)?
2. Define backtracking search in CSPs.
3. What is truth table.
4. Define propositional logic.
5. Define predicate logic.
6. What is the minimax algorithm in game playing?
7. What is a knowledge based agent?
8. Explain alpha beta pruning algorithm.
9. Describe the minimax algorithm for optimal decision making in games.
10. Describe the wumpus world as an example of a knowledge based agent environment.
11. Write a note on composite proposition.
12. Explain how propositional logic can be used for knowledge representation.
13. Explain Node coloring and Tic Tac Toe with an example.
14. Describe the process of effective inference rule in logical agent

UNIT – 3

1. Define the syntax of first order logic.
2. Write any two special symbols used in quantifier.
3. What is the difference between propositional logic and first order logic?
4. Define unification.
5. Define lifting.
6. Explain the resolution rule in first order logic and its application in theorem proving.
7. Explain the concept of quantifier with an example.
8. Discuss the unification algorithm and its role in first order logic inference. Provide example of unification.
9. Write a note on lifting.
10. Describe the forward chaining and backward chaining algorithm for inference in first order logic.
11. Explain the concept of wumpus world.

UNIT - 4

1. Define learning decision tree in machine learning.
2. What is ontological engineering in AI?
3. Define categories and objects in knowledge representation.
4. What are mental events and mental objects in AI?
5. Explain the concept of ontological engineering and its importance in AI.
6. Discuss the challenges and features in AI learning, including role of data quality.
7. Write a note on Internet shopping world.
8. Explain about actions, situation and events.
9. Explain the role of truth maintenance system.
10. Describe the process of learning decision tree.
11. Explain knowledge representation of learning and agent components.
12. Comparison of mental event and mental objects.
13. Write a short note on learning.

UNIT – 5

1. Define Robotics.
2. What are the main components of a robot?
3. What is application domain?
4. Define perception.
5. Define reactive control.
6. What is configuration space in motion planning?
7. Write a program to navigation in robotics.
8. Explain three layer architecture in robotics and its role in robotic software design.
9. Explain the challenges and opportunities of robotics hardware in different application domain.
10. Describe the different types of sensors used in robotics including their applications and limitations.
11. Explain the role of effectors in robotics, including the different types of effectors and their control systems.
12. Discuss the challenges of localization and mapping in robotics.
13. Describe the different types of perception in robotics.
14. Discuss the applications of robotics in different domains including manufacturing, healthcare and service robotics.
15. Describe the different methods for motion planning, including cell decomposition and skeletonization.
16. Describe the subsumption architecture and its role in robotic software design.

Course Content

UNIT: 1 – Introduction to AI, Informed Search and Exploration

10 Hours

Intelligent Agents: Agents and environment; the nature of environment; the structure of agents. Problem-solving: Problem solving agents; uninformed search strategies. Informed search strategies; Heuristic functions; On-line search agents.

UNIT: 2 - Constraint Satisfaction, Adversarial Search, Logical Agent

10 Hours

Constraint satisfaction problems; Backtracking search for CSPs. Adversarial search: Games; Optimal decisions in games; Alpha-Beta pruning. Knowledge-based agents; The wumpus world as an example world; Logic; propositional logic Reasoning patterns in propositional logic; Effective propositional inference; Agents based on propositional logic.

UNIT: 3 - First-Order Logic, Inference in First-Order Logic:

12 Hours

Syntax and semantics of first-order logic; Using first-order logic; Knowledge engineering in first-order logic. Propositional versus first-order inference; Unification and lifting; Forward chaining; Backward chaining; Resolution.

UNIT: 4 - Knowledge Representation and Learning, AI: Present and Future

10 Hours

Ontological engineering; Categories and objects; Actions, situations, and events; Mental events and mental objects; The Internet shopping world; Reasoning systems for categories; Reasoning with default information; Truth maintenance systems. Learning: Forms of Learning; Inductive learning; Learning decision trees.

UNIT: 5 – Introduction to Robotics :

10 Hours

Introduction; Robot Hardware: sensors and Effectors; Robotic Perception: localization, mapping, other types of perception; Planning to Move: configuration space, cell decomposition methods and skeletonization methods; Planning uncertain movements: robust methods; Moving: dynamics and control, potential field control and reactive control; Robotic Software: architectures, sub sumption architecture , three-layer architecture and robotic programming languages ; Application domains